

FRST vs Unified Coherence Dynamics - Comparison Report v1.0

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****Scope:**** compare the independently frozen Finite Relational Selection Theory (FRST) with Unified Coherence Dynamics (UCD) without silently merging either theory.

Executive result

FRST and UCD overlap strongly at the level of ****relational architecture****, but they are not the same theory.

FRST begins from a general mathematical object:

$$R_FRST = (K, H_raw, P_phys, A_phys, H, \rho_0, observables)$$

with a finite primitive carrier and the hierarchy

mathematical possibility $\rightarrow$ physical admissibility $\rightarrow$ realized state/history $\rightarrow$ observable reality.

UCD is much more specific. Its developed structures include the registration group

$$Q_72 = Z_3 \times Dic_6,$$

and the local operator space

$$H_504 = V_7 \text{ tensor } C[Q_72],$$

with $\dim(H_504)=7 \times 72 = 504$ and the developed $84+420$ sector structure.

The comparison therefore has a simple direction:

****FRST supplies a generic relational possibility architecture; UCD supplies a specific candidate realization and asks whether that realization can be forced from deeper source law.****

1. Possibility, admissibility and actuality

The generic comparison hierarchy is

$$\Omega_{\text{realized}} \subseteq \Omega_{\text{physical}} \subseteq \Omega_{\text{mathematical}}.$$

A convenient Hilbert-space form is

$$H_{\text{phys}} = P_{\text{phys}} H_{\text{raw}},$$

where P_{phys} is the orthogonal projector onto the common kernel/intersection of the physical consistency constraints.

The physically admissible state space is not the same thing as the actual state. The actual realized state/history requires additional data, schematically ρ_0 , an actualization law, or an equivalent global boundary condition.

This separates three questions that are often conflated:

1. What can be written mathematically?
2. What satisfies the physical law?
3. Which permitted state/history is actually realized?

2. Where FRST and UCD structurally agree

Both frameworks naturally organize physics around relational states rather than treating particles, fields, geometry and time as independent primitive substances.

Common structural themes include:

- physical states selected from a larger mathematical space;
- relational rather than label-dependent observables;
- local-frame/gauge consistency;
- closed-loop/holonomy information;
- a lawful Hamiltonian or equivalent dynamical generator;
- spectral excitation costs as a natural route to mass/gap structure;
- emergent or effective geometry from relational data;
- continuum mathematics as an effective description rather than automatically an ontological infinity;
- quantum-to-classical organization through stable records/observable sectors.

These similarities are architectural. They do not prove that FRST and UCD are mathematically equivalent.

3. Where UCD is more specific

FRST v1.0 does not independently derive the numbers 7, 72 or 504, nor the group Q_{72} , nor the H84/H420 split, nor the H-net construction.

UCD contains those specific candidate structures. In particular,

$$Q_{72} = Z_3 \times \text{Dic}_6,$$

$$H_{504} = V_7 \text{ tensor } C[Q_{72}],$$

and the finite ring/registration subsystem has the exact gap

$$\Delta_{504} = 2 \kappa \exp(-1/72).$$

At $\kappa=0.1$,

$$\Delta_{504} = 0.19724142334878325.$$

That number is a theorem of the defined finite H504/ring subsystem. It is not, by itself, the continuum Yang-Mills mass gap and it is not a theorem that the complete physical H-net is finite.

4. The critical non-import rule

FRST v1.0 starts with the axiom

$0 < |V| = N < \text{infinity}$.

UCD cannot simply inherit this as a theorem.

The UCD finite-realization branch is trying to establish why physically realized registration is finite from upstream UCD source principles. Therefore the comparison enforces:

FRST finite actuality axiom $\neq$ UCD finite-realization theorem.

This is the most important firewall in the comparison.

5. Completed creation is not yet a finiteness proof

A completed set may still be mathematically infinite. Therefore the statement that creation is finished does not, by itself, imply finite cardinality.

The UCD v14 finite-realization work isolated the missing source-level route. If the realized actualization structure is a rooted tree with finite roots, finite branching, and no infinite realized strict branch, then the realized tree is finite by Koenig's infinity lemma.

The two remaining upstream obligations are therefore:

- `P0-F1`: finite immediate lawful actualization / finite branching;
- `P0-F2`: no infinite physically realized strict differentiation chain / well-foundedness.

The implication finite roots + F1 + F2 $\Rightarrow$ finite realized tree is closed mathematics. F1 and F2 remain source theorems to derive inside UCD.

6. Selector theorem

FRST makes the selector explicit as P_{phys} . UCD already contains several physical-selection ingredients separately: registration, gauge, holonomy, provenance/subsystem structure and physical-sector constraints.

A useful UCD-level organizational definition is therefore

$$H_{\text{UCD}}^{\text{phys}} = \text{intersection_alpha } \ker(C_{\text{alpha}})$$

and

$$P_{\text{UCD}} = \text{Proj}(H_{\text{UCD}}^{\text{phys}}).$$

This definition collects existing constraints under one operator umbrella. It does not prove that the full UCD selector has been uniquely derived.

The deeper target is a source theorem of the form

Source law $\Rightarrow$ unique admissible selector $\Rightarrow$ Q72/H504/H-net architecture.

That theorem is open.

7. What the v2-v6 machine line established

v2 — selector, actuality and continuum envelope

The comparison was split into distinct questions: state selection, actual realization, finite realization and the relation between a finite physical ontology and a continuum mathematical approximation.

v3 — source-selector tournament

Finite-group averaging gives a canonical projector onto an invariant subspace, but symmetry alone does not generally select a unique theory. A finite alphabet plus bounded actualization arity gives finite branching, and a well-founded integer rank gives finite branch depth. These are conditional routes until UCD derives the required source premises.

v4 — native actualization grammar

Finite Q72/H504 basis labels provide a finite local codebook, but they do not prove global H-net finiteness. A finite-dimensional Hilbert space also contains continuously many pure-state rays, so finite basis-label count cannot substitute for a source-level finite actualization law.

v5 — finite separator basis

A finite complete separating witness family is sufficient to bound the number of distinguishable realized identities. This gives an exact operational route from a finite separator code to finite realized capacity.

v6 — route audit

If arbitrary predicates are allowed, a finite complete separator family exists exactly when the realized set is already finite. Therefore separator finiteness is a useful characterization/check, not the deeper explanation of why reality is finite. The source-dynamical burden returns to F1 and F2.

8. Continuum versus physical infinity

The comparison suggests a clean distinction:

- **finite physical ontology:** a finite set/network of actually realized primitive registrations;
- **continuum or infinite mathematical envelope:** a calculational/approximation structure used to express large-scale or limiting behavior.

A finite physical theory may still reproduce local continuum physics to high precision. If UCD establishes finite physical realization, older continuum constructions should therefore be interpreted carefully as effective or verification structures unless a stronger ontological claim is independently proved.

9. Integration classification

The first integration ledger uses these categories:

- ``READY_AS_DEFINITION`` — useful organizational notation that does not change UCD's physical content;
- ``ALREADY_STRUCTURALLY_PRESENT`` — FRST concept already represented by developed UCD structure;
- ``BRIDGE_REQUIRED`` — conceptual agreement exists but equivalence must be proved;

- `RECONCILIATION\_REQUIRED` — two existing formulations need an explicit compatibility theorem or interpretation;
- `DO\_NOT\_IMPORT\_AS\_THEOREM` — FRST assumes something UCD is still trying to derive.

The most important DO_NOT_IMPORT_AS_THEOREM entry is FRST finite actuality.

10. Current highest-value target

The comparison now points to one source-level question above the downstream applications:

****What property of the UCD source law forces finite, well-founded actualization and simultaneously restricts the wider relational possibility space strongly enough to produce the specific UCD selector?****

A successful result would need to establish, rather than assume, a chain resembling

Source/Law -> finite well-founded actualization -> physical selector -> Q72/H504/H-net -> observable physics.

Until that source chain is proved, the correct status is not “UCD and FRST are merged.” It is:

****FRST v1.0 is frozen independently; UCD is more specific; the comparison has exposed exact missing derivations and provides machines for testing candidate bridges.****

Reproducibility boundary

Machine PASS results in this repository mean that the corresponding encoded theorem, implication, countermodel or numerical identity replayed successfully. They are not experimental validation of either theory and do not turn conditional physical hypotheses into theorems.